CHANDIGARH UNIVERSITY
UTTAR PRADESH

Minor Project Progress Presentation
on
"AgroVision AI: AI-Powered Crop Disease Detection System"

Course : MCA(AIML)

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Topics

- Introduction of Project
- Problem Statement
- Project Objective
- Methodology
- Algorithms Used
- Technology Used
- System Architecture
- Conclusion

Introduction of Project

- AgroVision AI is an AI-powered crop disease detection system.
- It analyzes leaf images quickly using computer vision.
- It classifies the disease and shows prediction confidence scores.
- It gives expert-reviewed management guidance from a knowledge library.
- It supports early disease awareness for farmers.



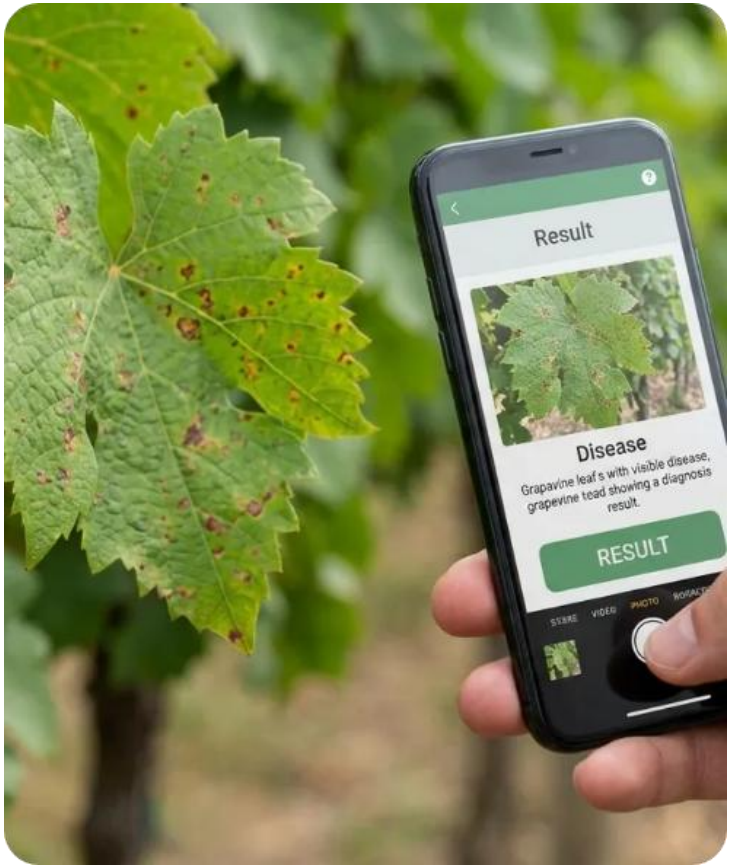
Problem Statement

- Manual visual diagnosis of crop diseases is complex and difficult.
- Non-disease stress symptoms look similar to real diseases.
- Diseases reduce overall crop yields and cause economic loss.
- Expert agricultural help is often delayed or hard to reach.
- Farmers need a fast, accessible tool for early disease awareness.



Project Objectives

- Develop a responsive browser-based web interface.
- Prepare and validate the PlantVillage image dataset.
- Train a MobileNetV2 image classification model.
- Build a fast real-time backend inference API.
- Show the predicted disease with a confidence score.
- Provide structured disease prevention and management guidance.



Proposed Methodology

1. **Dataset Preparation:** Validate the PlantVillage crop leaf image dataset.
2. **Preprocessing:** Resize and normalize the input images for the model.
3. **Model Training:** Train MobileNetV2 using deep transfer learning.
4. **Web Integration:** Connect the React frontend to the FastAPI backend through an API.
5. **Image Input:** User uploads a leaf image with secure file upload validation.
6. **Inference:** MobileNetV2 forward pass with softmax gives class probabilities.
7. **Result Delivery:** Display the predicted disease, confidence score and management guidance.

Algorithms Used

Deep Learning Techniques

1. CNN Architecture

- Extracts patterns from leaf images.
- Learns visual features of disease symptoms.

2. MobileNetV2

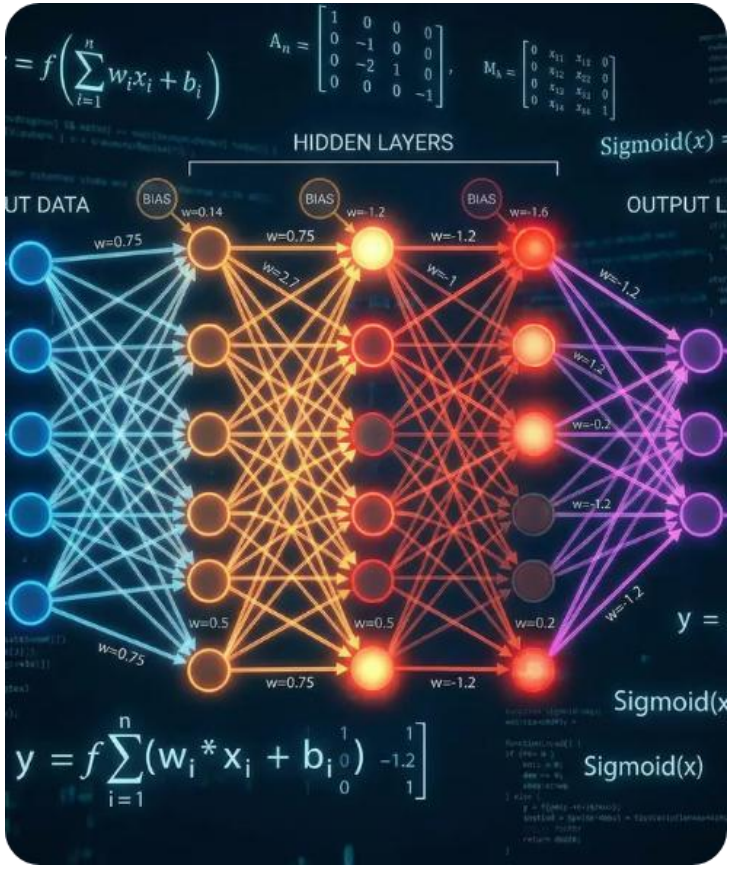
- Efficient, mobile-optimized feature maps.
- Lightweight model for fast inference.

3. Transfer Learning

- Adapts pre-trained network weights.
- Trained on PlantVillage leaf images.

4. Softmax Output

- Normalizes scores into class probabilities.
- Gives the prediction confidence score.



Technology Stack

- **Python:** Core language for the backend and model.
- **FastAPI:** Fast backend inference API framework.
- **MobileNetV2:** Image classification model.
- **PostgreSQL:** Relational database for persistence.
- **React:** Frontend web interface.
- **Tailwind CSS:** Responsive interface styling.
- **PlantVillage:** Leaf image dataset for training and validation.



System Architecture

Working Flow:

Image Input and Secure Upload Validation

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Preprocessing: Resize and Format Mapping

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Inference Engine: MobileNetV2 Prediction

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Result Delivery with Disease Guidance

Main Modules:

- Frontend: React and Tailwind CSS
- Backend API: Python and FastAPI
- Inference Engine: MobileNetV2
- Persistence: PostgreSQL Database
- Disease Guidance Knowledge Library

Conclusion

- AgroVision AI combines computer vision with expert-reviewed guidance.
- It classifies leaf diseases fast using MobileNetV2 transfer learning.
- It aims to be a scalable early awareness platform for farmers.

Future Scope

- Regional multilingual language support.
- Edge device offline inference.
- Transparent uncertainty threshold handling.
- Validated severity estimation features.
- Secure data management design.

Thank You!